

David Henderson, Ph.D.

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📍 San Francisco Bay Area

🔗 [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

SUMMARY

Ph.D. in Ecology, Evolution, and Population Biology with 6 years of experience in analyzing large-scale, complex ecological datasets, resulting in 4 peer-reviewed publications ([GoogleScholar](#)).

Ph.D.-trained expert in biodiversity, fire ecology and disturbance ecology. Strong track record of applying quantitative analysis and machine learning techniques, as demonstrated with AWS AI and Machine Learning certificates. Passionate about biogeography and using predictive modeling and interdisciplinary research in data-driven eco-evolutionary research.

TECHNICAL SKILLS

- **Domain Knowledge:** Forest ecology, fire ecology, disturbance ecology, landscape simulation, species distribution modeling, climate change impacts, biodiversity, carbon removal strategies
- **Data Analysis & Modeling:** R, Python, SQL, Linux, GIS, Tableau, STATA, SAS (multivariate analytics), AWS Cloud solutions, Google Analytics, machine learning, Earth-system modeling, species distribution modelling (SDM)
- **Project Management:** Stakeholder engagement, interdisciplinary collaboration, AGILE project management, quality control protocols

ACADEMIC HISTORY

- 2018–2023, **Ph.D.**, Ecology, Evolution, and Population Biology, Division of Biology and Biomedical Sciences, Washington University – St. Louis, U.S.A.
Dissertation title: *Biotic interactions and the maintenance of biodiversity across biogeographic gradients* ([link](#))
Committee: Dr. Jonathan Myers (advisor), Dr. J. Sebastian Tello, Dr. Rachel Penczykowski (chair), Dr. Brian E. Sedio, Dr. Adam B. Smith
- 2016, **B.Sc.**, Biological Sciences, Biology Department, Northern Illinois University, DeKalb, IL, U.S.A.
Summa cum laude

Ph.D. Research Biologist

2018—2023

Department of Biology, Washington University in St. Louis

- Successfully led projects making use of ‘omics and other data, developing methods to measure the impact of pathogens/natural enemies on tropical forest trees over an elevation gradient.

- Characterized chemical defensive mechanisms underlying species ranges in tropical forest trees using MS, bioinformatic (genetic, metabolomic), and network analysis techniques in lab and environmental chemical analyses in the field.
- Manuscript in revision at *Ecology*. Presented the results at the Botanical Society of America conference (Anchorage, AK).
- Co-organized and led an Andes-Biodiversity working group in English and Spanish with collaborators from South America, Europe and the United States that set the foundation for the above projects.

Visiting Faculty

2022

Dr. Brian Sedio Lab, University of Texas at Austin

- Selected as key collaborator with Sedio Lab to lead projects related to large scale metabolomic analyses and to develop the analysis pipeline using adapted genomic tools.
- Underwent intensive training focused on the analysis of plant metabolites including plant chemical-lab extractions, lab protocols, mass spectrometry alignment and analysis.
- Worked on updates to previously used bioinformatic pipelines and creation of original modeling approaches to define and compare chemical profiles (consisting of thousands of unique compounds) among individuals or groups.
- Worked with PI (Dr. Sedio), Lab Manager and undergraduate and high school students to contribute to the creation of training manuals, technical roadmaps, and documentation for lab protocols, mass spectrometry procedures, and metabolomic (bioinformatic) pipelines.

RECENT EMPLOYMENT

Software Developer

2024—Present

Galactic Advisors

- Utilized strict version control and development processes, contributing to the daily operations of quality control team.
- Maintained and created multi-structured database features. Optimized query rules for data transfer, analyses and automatic processes including data upload, preprocessing, formatting, analysis, and report generation.
- Updated most critical, most used sections of database schema within the first 30 days.
- Worked across security research team, software development team, and compliance team to produce deliverables within a short and urgent timeframe.
- Technical tools and proficiencies: CI/CD, Git, SQL, Python, Linux, Laravel, and Encryption management tools.

Research Program Mentor

2023 – Present

Polygence

- Work with dedicated high school students through completion of rigorous, self-directed research projects where they learn experimental design, scientific writing, data analysis, and a whole host of other skills.
- Maintain active communications with students and parents to ensure consistent progress throughout the projects.

- Simultaneously mentor 5 student research projects in ecology and environmental science, aiming at co-authoring in publications. See [mentor profile](#).

Biotech Consultant

2021—2023

The Biotechnology and Life Science Advising (BALSA) Group

- Worked as a project-based consultant, as a team member of a non-profit, student-led consulting group.
- Performed primary and secondary research including Surveys and Interviews with KPIs, Market Segmentation, Market Sizing, Competitive Analyses (SWOT), and Exit Strategies research for academic inventors, biotech innovators and investors, in a team-based environment.
- Stayed abreast of the latest developments and innovations, and stayed current on regulations and standards relevant to the project.

RELAVENT EXPERIENCE

PEER REVIEWED PUBLICATIONS

Published:

4. **Henderson, D.**, Sedio, B. E., Tello, J. S., Cayola, L., Fuentes, A. F., Alvestegui, B., Muchhala, N., & Myers, J. A. (2024). Ecological metabolomics of tropical tree communities across an elevational gradient: Implications for chemically-mediated biotic interactions and species diversity. (Accepted, Ecology).
3. Smith, A. B., Murphy, S. J., **Henderson, D.**, & Erickson, K. D. (2023). Including imprecisely georeferenced specimens improves accuracy of species distribution models and estimates of niche breadth. *Global Ecology and Biogeography*, 32(3), 342-355.
2. Rader, A. M., Cottrell, A., Kudla, A., Lum, T., **Henderson, D.**, Karandikar, H., & Letcher, S. G. (2020). Tree functional traits as predictors of microburst-associated treefalls in tropical wet forests. *Biotropica*, 52(3), 410-414.
1. Hazzi, N. A., Petrosky, A., Karandikar, H., **Henderson, D.**, & Jiménez-Conejo, N. (2020). Effect of forest succession and microenvironmental variables on the abundance of two wandering spider species (Araneae: Ctenidae) in a montane tropical forest. *The Journal of Arachnology*, 48(2), 140-145.

In Progress:

3. Chadwick, S.E., **Henderson, D.**, Cayola, L., Fuentes, A. F., Muchhala, N., Tello, J. S., Volf, M., Myers, J. A., Sedio, B. E. (2025). Chemical properties of foliar metabolomes represent a key axis of functional trait variation in forests of the tropical Andes. (Submitted to *PNAS*).

2. *Gomes, P. W., Mannocho-Russo, H., Schmid, R. ... (2024). plantMASST - Community-driven chemotaxonomic digitization of plants. (in revision Nature)*
1. **Henderson, D.**, Sedio, B. E., Tello, J. S., Cayola, L., Fuentes, A. F., Loza, M. I., Wu, Y. & Myers, J. A. (2024). Disentangling determinism and stochasticity in local tree neighborhoods across a tropical elevational gradient. (in prep for *Global Ecology and Biogeography*).

AWARDS AND GRANTS

- 2022: Botanical Society of America Travel Grant \$375
- 2022: American Society of Plant Taxonomists Graduate Student Grant \$500
- 2022: Washington University Division of Biology and Biomedical Science Graduate Student Travel Grant \$600
- 2022: Maxwell Hanrahan Foundation-Missouri Botanical Garden Funds (CO-PI), \$3,000
- 2021: Maxwell Hanrahan Foundation-Missouri Botanical Garden Funds (CO-PI), \$3,400
- 2020: ESA SEEDS Interdisciplinary Power of Data Research Travel Award Ecological Society of America \$500
- 2020: Living Earth Collaborative for “Testing the Role that Biotic Interactions Play in Shaping Elevational-Diversity Gradients: An Ecological Metabolomics Approach” (co-author), \$29,010
- 2020 – 2023: Catharine M. Lieneman Scholarship in Botany Washington University in St. Louis for full stipend amount

PRESENTATIONS

- 2022: Nov Biology Lab Talk – Metabolomic pipeline overview (for ecologists)
- 2022: Botanical Society of America Conference: Testing the role that biotic interactions play in shaping elevational-diversity gradients: An ecological metabolomics approach. Botanical Society of America Conference, Anchorage, AK.
- 2022: Jul Biology Lab Talk – Ecological Scale Metabolomics
- 2022: May BioForum – “Testing the Role that Biotic Interactions Play in Shaping Elevational Diversity Gradients: An Ecological Metabolomics Approach”.
- 2021: Canopy Science Forum – “Lonely Oaks, *Quercus humboldtii*”, virtual.
- 2020: Biology Department, University of Missouri - St. Louis, St. Louis, MO
- 2019: Ecological Society of America Conference: “Ecological Data Modeling for *Asclepias*”. Louisville, KY.
- April 2019 Midwest Ecology and Evolution Conference: “Ecological Data Modeling for *Asclepias*”. Terra Haute, IN.
- 2019: “Sapotaceae” for Anigosperm Phylogeny Plant Systematics group
- 2018: “Boraginales” for Anigosperm Phylogeny Plant Systematics group
- 2018: Danforth Microbe Meetup “Urban transects: Invaders as a prediction of infection”.

TEACHING EXPERIENCE

- 2022: Nominee Washington University in St. Louis Graduate Teaching Award
- 2021: Washington University Center for Teaching and Learning - Professional Development in Teaching Program completed

Teaching:

- 2020: Teaching Assistant, Community Ecology
- 2019: Teaching Assistant, Natural History of Missouri

Guest Lecturer:

- 2019–2023: Woody Plants of Missouri Systematics (undergraduate level), Washington University of St. Louis

Other Teaching Experience:

- 2022: Teaching Assistant, Vertical Voyage Tree-Climbing Class for K-12 Students, St. Louis, U.S.A

PROFESSIONAL DEVELOPMENT

- Google Analytics Academy: Google Analytics Certification (2025)
- Amazon Web Services: Certified Machine Learning – Specialty certification (2025)
- Amazon Web Services: Certified AI Practitioner certification (2024)
- Course Participant: Sample to Genome Course, Wellcome Sanger Institute, Hinxton, Cambridgeshire, UK (2024)
- 2021-2022: Bio Informatics and Lab Sample Training with Sedio Lab, University of Texas at Austin, Virtual - Austin, TX, U.S.A.
- 2022: Graphical Information Systems Applications Course – Environmental Studies 380, Washington University, St. Louis, MO, U.S.A.
- 2021: Power of Data Workshop, Ecological Society of America, Baltimore, MD, U.S.A.
- 2021: Tree Climbing Course, Vertical Voyages, St. Louis, MO, U.S.A.
- Washington University Wilderness First Aid practical course: Instructor Dr. Stanton Braude
- Professional Association of Diving Instructors (PADI) certified Rescue Diver

Selected Washington University BALSA Group seminars and workshops

- 2021: BALSA Summer Professional Development Series Session 1: How Research Gets from the Bench to the Patient Bedside, BALSA Group, St. Louis, MO, U.S.A.
- 2021: BALSA Summer Professional Development Series Session 2: Running Inclusive Meetings, BALSA Group, St. Louis, MO, U.S.A.
- 2021: BALSA Summer Professional Development Series Session 3: Business Communication, BALSA Group, St. Louis, MO, U.S.A.

- 2021: Balsa Summer Professional Development Series Session 4: Effective Slide Design for Business Meetings, Balsa Group, St. Louis, MO, U.S.A.
- 2020: Botany 2020 Conference Online Workshop, "Ace it! - Give a Better Talk" (Virtual)

Selected Washington University Center for Teaching and Learning seminars and workshops

- 2021: Creating a Teaching Portfolio, Developing Effective Summative Assessments
- 2020: Writing a Teaching Philosophy Statement, Engaging in Equity Pedagogy in the STEM Classroom, Crickets in the Classroom: Quick tips for Jumpstarting Conversation when Participation Lags
- 2019: Fostering an Inclusive Classroom Climate, Who's in Charge? Negotiating Your Role and Establishing Authority in the Classroom, Constructing Effective Collaborative Learning Opportunities, Using the Jigsaw Method to Increase Student Engagement
- 2019: Andes Working Group - Washington University, Tyson Research Center, Eureka, MO.
- 2019: Wilderness First Aid course – Washington University, Tyson Research Center, Eureka, MO.
- 2019, R Summer Workshop, Missouri Botanical Garden, St. Louis, MO, U.S.A
- 2018: Introduction to Pedagogical Scholarship Workshop, Planning and Organizing a Class Session, Facilitating Challenging Conversations, Motivating Student Learners
- 2018, Tropical Field course of Organization of Tropical Studies, Topic "Tropical Biology: An Ecological Approach," Costa Rica

SERVICE AND OUTREACH

- 2024 – Present: Washington University Bay Area Young Alumni Network, Co-Chair
- 2024: Volunteer CaveSim. San Mateo, CA
- 2023: Plant Sale: Harris World Ecology Center, University of Missouri in St. Louis
- 2023: Volunteer Festival of Nations, Missouri Botanical Garden, St. Louis, MO
- 2022: Plant Sale: Harris World Ecology Center, University of Missouri in St. Louis
- 2022: Panel Member - Washington University in St. Louis, Association of Black Biomedical Graduate Students
- 2022: Volunteer - Vertical Voyages, Columbia, MO
- 2021: Plant Sale: Harris World Ecology Center, University of Missouri in St. Louis
- 2020: Participant - Washington University in St. Louis
- 2020: Mentor - to incoming graduate students: Washington University in St. Louis, Initiative to Maximize Student Diversity
- 2019: Mentor - to incoming graduate students: Washington University in St. Louis, Initiative to Maximize Student Diversity

Referee:

- *Biotropica*, 2021
- *Oecologia*, 2020

SOCIETY AFFILIATIONS

- Washington University Young Alumni Network: Bay Area Co-Chair
- Ecological Society of America
- American Society of Plant Taxonomists
- American Society of Naturalists
- Botanical Society of America
- Association of Black Biomedical Graduate Students
- Washington University Student Union (WUGWU)
- Washington University Initiative to Maximize Student Diversity

LANGUAGES

English (fluent), Spanish (advanced, CEFR B2)

References:

- Dr. Jonathan Myers (jamyers@wustl.edu)
- Dr. Yingtong “Amanda” Wu (ytwu@stanford.edu)
- Dr. James Lucas (fiberethnobotanist@gmail.com)